

Structural Calculation Report

Reinforced Concrete Beam Design to BS EN 1992-1-1:2004+A1:2014 (Eurocode 2)

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Executive Summary

Overall Status:	FAIL	Max Utilisation:	$\eta = 123.5\%$
Section:	300.0×500.0 mm	Span:	6000.0 mm
Main Steel:	3-Ø16 (As=603.19 mm ²)	Links:	Ø8 @ 340.5 mm c/c (2 legs)

Recommendation: Provided steel insufficient. Use 1-Ø32 (As=804.2 mm², $\eta=80\%$).

1. Design Parameters

Section Type	Rectangular	
Web width (b_w)	300.0	mm
Overall depth (h)	500.0	mm
Nominal cover (c_{nom})	30.0	mm
Span (L)	6000.0	mm
Concrete grade	C30.0/37 ($f_{ck} = 30.0$ MPa)	
Steel yield strength (f_{yk})	500.0	MPa
Partial factor — concrete (γ_c)	1.5	
Partial factor — steel (γ_s)	1.15	
α_{cc}	0.85	

Applied Actions

Design bending moment (M_{Ed})	120.0	kNm
Design shear force (V_{Ed})	80.0	kN

2. Bending Design EN 1992-1-1 §6.1

Effective depth:

$$d = h - c_{nom} - \phi_{link} - \phi_{bar}/2$$

$$d = 500.0 - 30.0 - 8.0 - 16.0/2$$

$$\rightarrow 454.0 \text{ mm}$$

Design concrete strength:

$$f_{cd} = \alpha_{cc} \cdot f_{ck} / \gamma_c$$

$$f_{cd} = 0.85 \times 30.0 / 1.5$$

$$\rightarrow 17.00 \text{ MPa}$$

Design steel strength:

$$f_{yd} = f_{yk} / \gamma_s$$

$$f_{yd} = 500.0 / 1.15$$

$$\rightarrow 434.78 \text{ MPa}$$

K-factor:

$$K = M_{Ed} / (b_{eff} \cdot d^2 \cdot f_{ck})$$

$$K = 120.0 \times 10^3 / (300.0 \times 454.0^2 \times 30.0)$$

$$\rightarrow 0.0647 \text{ (} K' = 0.167 \text{)}$$

Lever arm:

$$z = d \cdot (0.5 + \sqrt{(0.25 - K/1.134)})$$

$$z = 454.0 \times (0.5 + \sqrt{(0.25 - 0.0647/1.134)})$$

$$\rightarrow 426.43 \text{ mm}$$

Required tension steel:

$$A_{s, req} = M_{Ed} / (f_{yd} \cdot z)$$

$$A_{s, req} = 120.0 \times 10^3 / (434.78 \times 426.43)$$

$$\rightarrow 647.24 \text{ mm}^2$$

Provided tension steel:

$$A_{s, prov} = n \cdot \pi \cdot \phi^2 / 4$$

$$A_{s, prov} = 3 \times \pi \times 16.0^2 / 4$$

$$\rightarrow 603.19 \text{ mm}^2$$

Effective depth (d)	454.0	mm
K-factor	0.0647	(K' = 0.167)
Neutral axis ratio (x/d)	0.152	
Lever arm (z)	426.43	mm
Required A _s	647.24	mm ²
Minimum A _{s,min}	205.14	mm ²
Provided A _{s,prov} (3-Ø16)	603.19	mm ²
Moment capacity (M _{Rd})	111.83	kNm
Utilisation ($\eta = M_{Ed}/M_{Rd}$)	107.3%	
Status	FAIL: INSUFFICIENT_STEEL	

3. Shear Design EN 1992-1-1 §6.2

Size factor:

$$k = \min(2.0, 1 + \sqrt{(200/d)})$$

$$k = 1 + \sqrt{(200/454.0)}$$

→ **1.664**

Concrete shear resistance:

$$V_{Rd,c} = [C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3}] \cdot b_w \cdot d$$

$$\rho_l = 0.00443$$

→ **64.40 kN**

Max strut capacity:

$$V_{Rd,max} = \alpha_{cw} \cdot b_w \cdot z \cdot v \cdot f_{cd} / (\cot\theta + \tan\theta)$$

$$\theta = 21.8^\circ, \cot\theta = 2.50$$

→ **446.36 kN**

Required shear reinforcement:

$$A_{sw/s} = V_{Ed} / (z \cdot f_{ywd} \cdot \cot\theta)$$

$$A_{sw/s} = 80.0 \times 10^3 / (408.6 \times 434.8 \times 2.50)$$

→ **0.263 mm²/mm**

V_{Ed}	80.0	kN
V_{Rd,c} (concrete resistance)	64.4	kN
V_{Rd,max} (strut capacity)	446.36	kN
Strut angle (θ)	21.8°	(cotθ = 2.5)
Required A_{sw}/s	0.263	mm ² /mm
Max link spacing (s_{max})	340.5	mm
Provided: Ø8 @ 340.5 mm	2 legs	
Utilisation (η)	61.0%	
Status	OK	

4. Deflection Check EN 1992-1-1 §7.4.2

Reference reinforcement ratio:

$$\rho_{\min} = \sqrt[3]{f_{ck}} \times 10^{-3}$$

$$\rho_{\min} = \sqrt[3]{30.0} \times 10^{-3}$$

→ **0.00548**

Actual reinforcement ratio:

$$\rho = A_{s,req} / (b \cdot d)$$

$$\rho = 647.24 / (300.0 \times 454.0)$$

→ **0.00475**

Basic span/depth ratio:

$$L/d_{basic} = K \cdot [11 + 1.5\sqrt{f_{ck}} \cdot \rho_{\min} / \rho + \dots]$$

$$K_{sys} = 1.0$$

→ **21.51**

Modification factor:

$$\text{Factor} = (500/f_{yk}) \cdot (A_{s,prov}/A_{s,req}) \leq 1.5$$

$$\text{Factor} = (500/500.0) \times (603.2/647.2)$$

→ **0.932**

Allowable span/depth:

$$L/d_{allow} = L/d_{basic} \times \text{Factor}$$

$$21.51 \times 0.932$$

→ **20.05**

Actual span/depth:

$$L/d_{actual} = L / d$$

$$6000.0 / 454.0$$

→ **13.22**

Actual L/d	13.22	
Allowable L/d	20.05	
Utilisation (η)	65.9%	
Status	OK	

5. Cracking Control EN 1992-1-1 §7.3.3

Estimated steel stress:

$$\sigma_s = 310 \cdot (f_{yk}/500) \cdot (A_{s,req}/A_{s,prov})$$

$$\sigma_s = 310 \times (500.0/500) \times (647.2/603.2)$$

→ **332.6 MPa**

Max allowable spacing:

From Table 7.3N for $w_k = 0.3$ mm

$$\sigma_s = 332.6 \text{ MPa} \rightarrow \text{interpolate}$$

→ **84.2 mm**

Actual bar spacing:

$$s = (b - 2c - 2\phi_{link} - n \cdot \phi_{bar}) / (n-1) + \phi_{bar}$$

$$n = 3, \phi = 16.0 \text{ mm}$$

→ **104.0 mm**

Estimated steel stress (σ_s)	332.6	MPa
Actual bar spacing	104.0	mm
Max allowable spacing (Table 7.3N)	84.2	mm
Crack width limit (w_k)	0.3	mm
Utilisation (η)	123.5%	
Status	FAIL	

6. Utilisation Summary

Check	Utilisation (η)	Status
Bending (§6.1)	107.3%	✗
Shear (§6.2)	61.0%	✓
Deflection (§7.4)	65.9%	✓
Cracking (§7.3)	123.5%	✗
Maximum Utilisation	123.5%	FAIL

Assumptions & Limitations

- Simply supported beam with uniform loading assumed for deflection check.

- Concrete properties derived from Table 3.1 of EN 1992-1-1.

- Exposure class: XC1 (unless otherwise specified). Minimum cover per Table 4.4N.

- Shear design uses Variable Strut Inclination Method per §6.2.3.

- Deflection checked using simplified span/effective depth method per §7.4.2 (Table 7.4N).

- Cracking control via bar spacing rules per §7.3.3 (Table 7.3N). Direct crack width calculation not performed.

- Quasi-permanent load ratio assumed as 0.6 for steel stress estimation in cracking check.

- Fire resistance, durability detailing, and anchorage/lap lengths are outside the scope of this calculation.

- This report is for preliminary design purposes. Independent checking is recommended for final design.